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Control Your AI Mind

Instructor Guide

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This guide is for teaching the course live. It assumes the complete training manual is the reference document and the applied lab book is the hands-on companion.

Recommended Formats

Format A: 8-Week Course

- Week 1: Control, not outsource
- Week 2: Setup, permissions, sandbox
- Week 3: Task contracts
- Week 4: Data workflow I: folder to figure
- Week 5: Data workflow II: scrape to database / large data
- Week 6: Writing, thinking, and skills
- Week 7: Verification and econometrics checklists
- Week 8: Research-agent OS capstone

Format B: 2-Day Intensive

- Day 1 morning:** Control, setup, task contracts.
- Day 1 afternoon:** Folder-to-figure and scrape-to-database labs.
- Day 2 morning:** Writing, skills, memory, and verification.
- Day 2 afternoon:** Capstone OS build and presentations.

Format C: 1-Day Executive Workshop

Focus on M1, M2, M3, M5, M7, and a simplified M8. Avoid deep DuckDB/HMDA implementation unless the audience is technical.

Teaching Principles

1. **Start from a real research problem.** Students learn less from toy prompts and more from their own messy project.
 2. **Make every artifact inspectable.** If a figure is shown, the class should know which script made it and which data it used.
 3. **Do not let students prompt too early.** First define task, source, output, and checks.
 4. **Separate tool fluency from research control.** Tool fluency matters, but only after control discipline is in place.
 5. **Use failures as teaching material.** 403 errors, bad package versions, broken joins, and hallucinated citations are the lesson.
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Session-Level Plans

Session M1: Control, Not Outsource

Goal. The learner must treat AI as an operator inside a research production function, not as a substitute for research judgment.

90-minute structure.

- 0-10 min: opening case and diagnostic question.
- 10-25 min: concept lecture.
- 25-45 min: source-backed demo.
- 45-65 min: guided student build.
- 65-80 min: peer review / failure analysis.
- 80-90 min: deliverable checklist and homework.

Board prompts.

- What is the agent allowed to decide here?
- What would a false success look like?
- Which evidence would make you believe the output?
- What should be saved to disk before the session ends?

Instructor warnings.

- Do not reward speed without traceability.
- Do not let students accept model-written interpretations without claim audit.
- Do not let tool setup consume the full class; have a fallback demo ready.

Homework. Control map with non-delegable decisions, delegation rules, and verification checkpoints.

Session M2: Setup, Permissions, Sandboxes

Goal. Agentic work begins before the prompt: the environment determines what the agent can see, touch, break, leak, or verify.

90-minute structure.

- 0-10 min: opening case and diagnostic question.
- 10-25 min: concept lecture.
- 25-45 min: source-backed demo.
- 45-65 min: guided student build.
- 65-80 min: peer review / failure analysis.
- 80-90 min: deliverable checklist and homework.

Board prompts.

- What is the agent allowed to decide here?
- What would a false success look like?
- Which evidence would make you believe the output?
- What should be saved to disk before the session ends?

Instructor warnings.

- Do not reward speed without traceability.
- Do not let students accept model-written interpretations without claim audit.
- Do not let tool setup consume the full class; have a fallback demo ready.

Homework. AGENTS.md/CLAUDE.md-style instruction pack plus environment risk checklist.

Session M3: Task Contracts For Research Agents

Goal. A good task contract turns vague intent into inputs, outputs, constraints, checks, and stopping rules.

90-minute structure.

- 0-10 min: opening case and diagnostic question.
- 10-25 min: concept lecture.
- 25-45 min: source-backed demo.
- 45-65 min: guided student build.
- 65-80 min: peer review / failure analysis.
- 80-90 min: deliverable checklist and homework.

Board prompts.

- What is the agent allowed to decide here?
- What would a false success look like?
- Which evidence would make you believe the output?
- What should be saved to disk before the session ends?

Instructor warnings.

- Do not reward speed without traceability.
- Do not let students accept model-written interpretations without claim audit.
- Do not let tool setup consume the full class; have a fallback demo ready.

Homework. Two reusable task contracts with acceptance checks.

Session M4: Data Workflows: Folder To Figure To Database

Goal. The practical value of AI agents is clearest when source collection, parsing, schema design, validation, and output generation are made explicit.

90-minute structure.

- 0-10 min: opening case and diagnostic question.
- 10-25 min: concept lecture.
- 25-45 min: source-backed demo.
- 45-65 min: guided student build.
- 65-80 min: peer review / failure analysis.
- 80-90 min: deliverable checklist and homework.

Board prompts.

- What is the agent allowed to decide here?
- What would a false success look like?
- Which evidence would make you believe the output?
- What should be saved to disk before the session ends?

Instructor warnings.

- Do not reward speed without traceability.
- Do not let students accept model-written interpretations without claim audit.
- Do not let tool setup consume the full class; have a fallback demo ready.

Homework. Data workflow plan, schema, validation checklist, and one reproducible figure/table.

Session M5: Writing And Thinking With AI

Goal. AI should improve writing clarity without stealing the thinking work that makes a researcher accountable.

90-minute structure.

- 0-10 min: opening case and diagnostic question.
- 10-25 min: concept lecture.
- 25-45 min: source-backed demo.
- 45-65 min: guided student build.
- 65-80 min: peer review / failure analysis.
- 80-90 min: deliverable checklist and homework.

Board prompts.

- What is the agent allowed to decide here?
- What would a false success look like?
- Which evidence would make you believe the output?
- What should be saved to disk before the session ends?

Instructor warnings.

- Do not reward speed without traceability.
- Do not let students accept model-written interpretations without claim audit.
- Do not let tool setup consume the full class; have a fallback demo ready.

Homework. Writing style guide, claim audit, and revised paragraph.

Session M6: Skills, Memory, Reusable Methods

Goal. A skill is not a power-up; it is a reusable encoding of how the researcher wants recurring judgment to be performed.

90-minute structure.

- 0-10 min: opening case and diagnostic question.
- 10-25 min: concept lecture.
- 25-45 min: source-backed demo.
- 45-65 min: guided student build.
- 65-80 min: peer review / failure analysis.
- 80-90 min: deliverable checklist and homework.

Board prompts.

- What is the agent allowed to decide here?
- What would a false success look like?
- Which evidence would make you believe the output?
- What should be saved to disk before the session ends?

Instructor warnings.

- Do not reward speed without traceability.
- Do not let students accept model-written interpretations without claim audit.
- Do not let tool setup consume the full class; have a fallback demo ready.

Homework. Reusable skill/checklist plus two test runs.

Session M7: Verification Loops And Research O-Rings

Goal. The more agents produce, the more deliberate verification must become. Automation does not eliminate domain human capital; it raises its returns.

90-minute structure.

- 0-10 min: opening case and diagnostic question.

- 10-25 min: concept lecture.
- 25-45 min: source-backed demo.
- 45-65 min: guided student build.
- 65-80 min: peer review / failure analysis.
- 80-90 min: deliverable checklist and homework.

Board prompts.

- What is the agent allowed to decide here?
- What would a false success look like?
- Which evidence would make you believe the output?
- What should be saved to disk before the session ends?

Instructor warnings.

- Do not reward speed without traceability.
- Do not let students accept model-written interpretations without claim audit.
- Do not let tool setup consume the full class; have a fallback demo ready.

Homework. Verification checklist and adversarial review memo.

Session M8: Build Your Research Agent OS

Goal. The end product is not a favorite AI tool; it is a controlled research operating system: sources, tasks, skills, memory, tools, and checks.

90-minute structure.

- 0-10 min: opening case and diagnostic question.
- 10-25 min: concept lecture.
- 25-45 min: source-backed demo.
- 45-65 min: guided student build.
- 65-80 min: peer review / failure analysis.
- 80-90 min: deliverable checklist and homework.

Board prompts.

- What is the agent allowed to decide here?
- What would a false success look like?
- Which evidence would make you believe the output?
- What should be saved to disk before the session ends?

Instructor warnings.

- Do not reward speed without traceability.
- Do not let students accept model-written interpretations without claim audit.
- Do not let tool setup consume the full class; have a fallback demo ready.

Homework. Capstone research-agent OS map plus one completed workflow.

Grading Rubrics

Task Contract Rubric

- 5: Goal, inputs, outputs, non-goals, assumptions, checks, and stop conditions are explicit.
- 4: Most fields present; one or two checks are vague.
- 3: Task is understandable but not strongly bounded.
- 2: Prompt-like request with weak acceptance criteria.
- 1: Vague instruction; no verification path.

Data Workflow Rubric

- 5: Raw data preserved, scripts reproducible, metadata documented, validation checks meaningful, output traceable.
- 4: Workflow reproducible but metadata or sanity checks incomplete.
- 3: Output exists but several steps rely on implicit decisions.
- 2: Manual steps or opaque transformations dominate.
- 1: Output cannot be reproduced.

Writing Audit Rubric

- 5: Every claim is labeled and tied to evidence; overclaims are corrected.
- 4: Most claims checked; some support remains broad.
- 3: Good prose edits but weak claim discipline.
- 2: Model rewrote text without preserving evidence trail.
- 1: Fluent but unsupported writing.

Verification Memo Rubric

- 5: Identifies likely failure modes, checks them directly, and reports residual uncertainty.
 - 4: Checks are good but not adversarial enough.
 - 3: Basic checks only.
 - 2: Verification is mostly narrative.
 - 1: No real verification.
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Slide Deck Outline**Deck 1: Control**

1. Title and course promise
2. Why prompt tricks are not enough
3. Research as an O-ring production process
4. The human-agent division of labor
5. Delegation ladder
6. False success examples
7. Control map template
8. Exercise instructions

Deck 2: Setup

1. Tool ladder
2. Context window as working memory
3. Session state belongs on disk
4. Folder permissions vs tool permissions
5. Sandbox-first principle
6. Sensitive data policy
7. Repo setup template
8. Exercise instructions

Deck 3: Task Contracts

1. Bad prompt vs good contract
2. Contract fields
3. Non-goals and stopping rules
4. Ask-before-acting policies
5. Acceptance checks
6. Three conversion examples
7. Student contract build

Deck 4: Data Workflows

1. From question to source

2. Raw data preservation
3. Folder-to-figure pipeline
4. Scrape-parse-structure-analyze
5. DuckDB/parquet/metadata
6. Sanity checks
7. Lab instructions

Deck 5: Writing And Skills

1. Writing as thinking
2. Cognitive offloading
3. Comments-only editing
4. Style guide as constraint
5. Skills as repeated judgment
6. Skill template
7. Lab instructions

Deck 6: Verification

1. Production vs verification
 2. Zero-error constraint
 3. DiD/CSDID example
 4. Psychological verification
 5. Adversarial review loops
 6. Verification memo template
 7. Capstone instructions
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Instructor Operating Checklist

Before class:

- Prepare a clean demo repo.
- Prepare one intentionally flawed output.
- Prepare fallback screenshots if live tool/API access fails.
- Check that no private corpus full text is displayed publicly.
- Have a source map and task contract template ready.

During class:

- Force students to write the task before using the agent.
- Pause after every model output and ask what would make it wrong.
- Save state to files during demos.
- Show at least one failure or correction.

After class:

- Collect artifacts, not chat logs.
- Grade traceability and verification.
- Update source atlas with new student-supplied materials.